Appendix E
Biophysics of the Human Body

This appendix places the contents of this text within the field of *biophysics*.

Biophysics is hard to define well, as is illustrated by the many definitive, yet different definitions of biophysics provided in [1–12]. Broadly speaking, biophysics is the applications of physics and physical principles to biology. In this context, virtually everything presented in this book is biophysics. However, this term is often used in the more restricted sense of the use of physics at a more molecular and cellular level. We will use this narrower context for the rest of this appendix, and in this restricted sense many topics covered here are still biophysics, but many areas in biophysics have not been covered. Yet another definition of biophysics is the study of biology using physical methods. This is distinguished from *biological physics*, which is the study of the physical properties of biology.

One topic in biophysics is the molecular structure of biological systems. This includes the electrostatics of ions in solutions (Chap. 12), the structure of biomacromolecules, such as proteins and areas such as protein folding, structure, and properties of interfaces between biological media such as cell membranes (surface tension in Chaps. 7 and 9, nerve cell membranes in Chap. 12), and ion channels in membranes (which is very briefly touched in nerve conduction in Chap. 12).

Statistical mechanics is the examination of systems composed of many similar objects or systems, each of which is well characterized. The whole ensemble of systems often is in thermal equilibrium and, after statistical mechanical analysis, can be treated by using thermodynamics. The treatment of ions in solution in Chap. 12 is the result of statistical mechanics. As stated in Chap. 5, the Hill force–velocity curve of muscles can be derived from statistical mechanical analysis of the many actin–myosin crossbridges (which was not done here). Many aspects of cell membranes and protein structure—such as folding—can be examined by using statistical mechanical methods.

Biophysics includes the bioenergetics of the photosynthesis process and the synthesis of ATP and its use. In Chap. 6 we examined the biophysics of energy usage in the human body. The movement of organisms, such as bacteria motion and muscular movement, are part of molecular and cellular biophysics; the microscopic basis of muscle operation was explored in Chap. 5. The electrochemical properties of cell

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membranes and nerve signals, as discussed in Chap. 12, have always been central topics in biophysics. Some include within biophysics the higher-level integration and combinations of molecular and cellular systems, such as memory, control of movement, visual integration, and consciousness and thinking.

The use of physical characterization to biological problems plays a central role in biophysics, such as the use of X-ray diffraction (XRD) to determine molecular structure, nuclear magnetic resonance (NMR) to study molecules in more natural environments than X-ray diffraction can be used, scanning tunneling microscopy (STM) to examine the atomic structure of surfaces, atomic force microscopy (AFM) to examine surfaces and to measure forces, and optical tweezers to manipulate molecules. Both AFM and optical tweezers have been instrumental in studying the fundamental interactions in muscles, such as individual actin–myosin crossbridges (Chap. 5).

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